

Dam Hong Phuc

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Education

FPT University, Ha Noi
B.Tech in Artificial Intelligence

Sep 2022 - Dec 2025
GPA: 8.49/10

Professional Experience

FPT Software

AIops & Platform Engineer

January 2026 – Present

- **Project: Enterprise HR Conversational AI Platform – Covestro AG**
 - * Owned a production HR assistant for a Fortune 500 client (**LangGraph** multi-agent, **Bedrock Knowledge Bases**, **OpenSearch Serverless**, 9 languages) as sole developer and led its **zero-downtime migration** from a legacy monolith to an event-driven platform of **MCP** services (**ServiceNow**, **Workday**, knowledge) on **AWS**.
 - * Led the **customer-facing** engagement as the client's trusted technical advisor: worked directly with the Product Owner and HR business users, drove **root-cause analysis**, presented options with cost/risk trade-offs, ran **UAT**, and secured security and architecture sign-off for every production change.
 - * Designed multi-layer **metadata filtering** (country, legal entity, HR-only tier) resolved **server-side per turn**, fail-closed at a single retrieval choke point, and shipped the restricted **HR4HR** knowledge base with no leakage window; added a permission-scoped **semantic cache** on **ElastiCache** serving repeat questions in ~150 ms.
 - * Re-architected the platform from v1 to v2 end to end: agent-as-config **LangGraph** engine with **MCP** services, traffic cutover through a fail-closed **SSM** switch with rehearsed rollback, stateful resources moved across **CDK** apps via **CloudFormation stack refactor + cdk import** with zero id changes, and **CI/CD** rebuilt on **GitHub Actions OIDC**.
 - * Architected the security posture for a regulated enterprise: **Bedrock Guardrails** with real-time **PII redaction**, VPC-private Lambdas under least-privilege cross-account **IAM/KMS**, and **Langfuse** observability with content dropped in production.
- **Project: Enterprise AI Contract Intelligence System**
 - * Upgraded the legacy system to a **Skill-based framework** to parse unstructured contracts. Employed prompt chaining and vector **RAG (Qdrant)** to slash hallucinations and boost extraction accuracy by 35%.
 - * Dynamically routed multi-model traffic via **AWS Bedrock** to scale for thousands of documents. Synchronized latency via **SSE streaming**.
 - * Executed end-to-end migration from local deployments to an enterprise-grade **AWS** infrastructure to scale heavy batch algorithms horizontally.

HTI Group

AI Engineer & LLMops

April 2025 – December 2025

- **Project: Enterprise Agentic Chatbot & LLMops Infrastructure** 
 - * Rapidly prototyped **Multi-Agent** solutions with **LangGraph** and **LlamaIndex** following **ReAct Pattern** to unify institutional knowledge. Orchestrated cloud-native microservices using **AWS Lambda**, **ECS Fargate**.
 - * Automated **CI/CD pipelines** via IaC (**AWS CDK/TypeScript**) for repeatable deployments. Implemented distributed observability with **Langfuse** to trace agent trajectories and ensure hallucination-free outputs.
 - * Designed a **Data Ingestion Pipeline** to normalize multi-format sources (Excel, Docs, MD, PPT, PNG, CSV). Applied **Hierarchical Semantic Chunking** with **Auto Merge** and **LLM Contextualization** for chunk chunks.
 - * Combined diverse retrieval strategies: **Vector Search**, **Full-text Search**, **Graph Search**, and **Pandas Retrieval**; integrated tool calling to external services (Discord, Google Calendar).

HG Group

AI Engineer

May 2023 – March 2025

- **Project: HG Chatbot - Enterprise Conversational AI**
 - * Engineered robust data pipelines to parse and normalize unstructured content across diverse formats (PDF, Docx, PPT, Excel, PNG). Architected an advanced search engine utilizing **Hybrid Retrieval** and **Cross-Encoder reranking** to maximize context relevance.
- **Project: HG Render - AI Media Recommendation Engine**
 - * Architected a recommendation engine over 1M+ multimodal assets (images, videos, SFX) using dense retrieval (**Qdrant**) and **GPT** for combinatorial optimization to assemble production-ready videos.
 - * Engineered async background task processing with **Celery**, **Redis**, and **asyncio** to handle long-running video assembly jobs at scale.
- **Project: HIG (HG Image Generation) - Internal AI Asset Creation**
 - * Translated SOTA text-to-image diffusion papers into operational internal workflows. Engineered robust visual data pipelines to curate and preprocess large-scale datasets, establishing high-quality training baselines.
 - * Fine-tuned **FLUX.1** models with LoRA to highly specific industry design styles. Managed the generation of **custom visual assets**  (**Lofi-cartoon aesthetics**) to drive enterprise media campaigns.

Startup Product

Wizzdom

July 2025 – December 2025

Technical Leader & Founding Member

- Architected an AI-driven EdTech platform targeting Math educators. The platform transforms inefficient manual assessment habits into a 10x faster workflow, delivering personalized student feedback and eliminating hours of repetitive grading.
- Spearheaded end-to-end development, rapidly prototyping and pivoting iterations to accelerate product-market fit. Engineered complex AI evaluation pipelines using **Gemini** models and successfully managed the scalable cloud deployment on **AWS**.

Technical Skills

GenAI on AWS: Bedrock (Knowledge Bases, Guardrails, Rerank, Claude & Cohere models), OpenSearch Serverless, Lambda, ECS Fargate, SQS, EventBridge, DynamoDB, ElastiCache, S3, RDS, IAM/KMS, VPC, CDK, CloudFormation

Agentic & RAG: LangGraph (ReAct, multi-agent), MCP, LangChain, LlamaIndex, prompt/context engineering, hybrid search, reranking, hierarchical chunking, semantic caching, Qdrant, Neo4j, PostgreSQL, MongoDB

FM Evaluation & LLMops: RAGAS, LLM-as-judge, recall@k harnesses, A/B parity testing, Langfuse, LangSmith, PII redaction, GitHub Actions CI/CD, Docker, Kubernetes, Ray

Fine-tuning & Engineering: LoRA/QLoRA (FLUX.1), vLLM, PyTorch, Python (FastAPI, gRPC, async), TypeScript, event-driven microservices, ServiceNow REST API integration, Kafka, RabbitMQ

Certifications

AWS Certified Generative AI Developer – Professional (AIP-C01) – Amazon Web Services, 2026

AWS Certified Solutions Architect – Associate (SAA-C03) – Amazon Web Services, 2026

Deep Learning Specialization – DeepLearning.AI (Andrew Ng), 2024

IBM Full Stack Software Developer Professional Certificate – IBM, 2024

Software Development Lifecycle Specialization – University of Minnesota, 2024

IELTS 7.0 – English proficiency

Full list with credentials: github.com/maver1ch/Certifications